

Image Captioning in the Wild: An Empirical Study

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Abstract

Image captioning serves as a crucial bridge between visual perception and natural language understanding, essential for real-world applications such as accessibility tools for the visually impaired. Developing robust models for in-the-wild data such as VizWiz exposes critical challenges in image quality and domain shift. In this work, we present a systematic, empirical evaluation of image captioning architectures ranging from baseline frameworks to state-of-the-art foundation models. We evaluate architectures spanning from basic Encoder-Decoder pipelines—isolating backbones, decoders, and attention mechanisms—to pre-trained VLMs (BLIP) and modern MLLMs (Qwen-VL), via zero-shot inference and LoRA fine-tuning. Our results demonstrate that while basic models benefit significantly from pre-aligned features and adaptive attention, overcoming severe domain shifts strictly requires end-to-end joint fine-tuning. While the Qwen3.5-based model delivers superior zero-shot semantic richness, adapting it via linear projectors induces modality collapse, causing the model to ignore visual inputs entirely. Ultimately, our study highlights the critical trade-offs between architectural complexity, pre-training alignment, and domain adaptability.

1. Introduction

Image captioning is the task of automatically generating natural language descriptions for images. It serves as a fundamental bridge between Computer Vision (CV) and Natural Language Processing (NLP). Unlike standard classification or object detection, which output discrete labels or bounding boxes, captioning requires a system to recognize objects, understand their spatial and semantic relationships, and articulate this understanding into a syntactically correct, coherent sequence of words.

The translation from pixels to text is critical for numerous real-world applications. Primarily, high-quality captioning systems are deployed in accessibility tools, enabling

screen readers to describe digital environments and web images to visually impaired users [5, 9, 13]. Furthermore, it is a core component in content-based image retrieval, allowing users to search for unannotated images using complex text queries [12, 18, 21], and in automated metadata generation for large-scale media databases [43]. In robotics and embodied AI, captioning facilitates human-robot interaction by allowing machines to communicate their visual perception of dynamic environments through natural language [33].

Over the last decade, architectures driving image captioning have shifted dramatically, from early encoder-decoder networks to pipelines leveraging independently pretrained vision-language components, and currently to modern Multimodal Large Language Models (MLLMs). While performance on standard benchmarks has continuously improved, the rapid progression often obscures the specific impact of individual architectural choices. As models grow in scale and complexity, the isolated contributions of different encoders, decoders, attention mechanisms, and training paradigms become difficult to decouple.

In this work, we present an empirical study designed to deconstruct the evolutionary trajectory of image captioning models. Rather than proposing new architectures, we systematically evaluate and compare existing models across the historical spectrum to understand the exact behavior and trade-offs of their core components under a unified experimental setup. All experiments are conducted on the VizWiz dataset [14], which provides a challenging real-world benchmark with images captured by visually impaired users. Specifically, our contributions are threefold:

- **Systematic Baseline Deconstruction:** We provide a granular analysis of classic encoder-decoder frameworks, isolating the impact of visual backbones (Convolutional Neural Networks, CNNs [22], and Vision Transformers, ViTs [11]), recurrent decoders (including the recent Extended Long Short-Term Memory, xLSTM [4]), tokenization strategies (character, subword, and word-level), and attention mechanisms to establish robust baselines.

- **Evaluation of Vision-Language Model (VLM) Adaptation Strategies:** We transition to pretrained VLMs (Bootstrapping Language-Image Pretraining, BLIP [23], and ViT+Generative Pretrained Transformer 2, GPT-2 [31]), specifically evaluating how different fine-tuning configurations and modality-freezing strategies impact cross-modal alignment and data efficiency.
- **Analysis of MLLM Scaling & Fine-Tuning Dynamics:** We assess modern MLLMs (the Qwen-VL family) in both zero-shot settings and under Parameter-Efficient Fine-Tuning (PEFT) constraints, identifying key adaptation challenges such as the modality collapse phenomenon in lightweight architectures.

2. Related Work

We organize our review along three axes that contextualize our approach: (i) encoder-decoder models with attention mechanisms, (ii) pretrained vision-language models, and (iii) Multimodal Large Language Models.

Basic Image Captioning Framework. Neural approaches to image captioning began with multimodal log-bilinear models [20] and multimodal recurrent networks [29], demonstrating the potential of learning joint visual-linguistic representations. Inspired by sequence-to-sequence models for machine translation [7, 35], Vinyals *et al.* [39] proposed *Show and Tell*, consolidating the encoder-decoder paradigm: a CNN pretrained on ImageNet encodes images into fixed-length vectors, and an LSTM decodes them into captions word by word. A key limitation of this formulation is the compression into a single vector, causing the decoder to overlook fine-grained spatial details. Xu *et al.* [42] addressed this with *Show, Attend and Tell*, introducing soft and hard attention mechanisms that allow the decoder to selectively focus on different spatial regions at each generation step. Soft attention computes a weighted average over all spatial locations using a learned alignment function, while hard attention samples a single region and is trained via REINFORCE [41]. Lu *et al.* [28] further refined this paradigm with an *adaptive attention* model that uses a visual sentinel. While these classical models laid the foundation for modern captioning, systematically isolating the individual impact of their components (encoder, decoder, attention, and hyperparameters) remains unexplored.

Pretrained Vision-Language Models. The success of pretrained language models such as BERT [10] and GPT-2 motivated leveraging pretrained components for captioning. A natural approach pairs a pretrained vision encoder (*e.g.* ViT) with a pretrained language decoder (*e.g.* GPT-2) and fine-tunes the pipeline for captioning. However, the encoder and decoder are pretrained on separate objectives and lack cross-modal alignment. Li *et al.* [23] addressed

this with BLIP, which jointly pretrains visual and language components using contrastive, matching, and generation objectives, together with a captioner-filter strategy that cleans noisy web data, producing stronger cross-modal representations than independently pretrained pipelines. Despite these advances, the granular impact of selective fine-tuning strategies, such as the effects of freezing versus unfreezing specific modalities, requires further investigation to optimize training efficiency and performance.

Multimodal Large Language Models. The most recent paradigm integrates visual perception directly into LLMs, yielding models capable of open-ended image understanding beyond fixed-vocabulary captioning. Models such as LLaVA [26], InternVL [6], and Qwen-VL [1–3, 40] share a common architectural template: a ViT encoder, a cross-modal projector, and a pretrained LLM backbone. Advances in dynamic resolution and positional encoding have allowed open-weight models like Qwen2-VL to approach the performance of proprietary systems. However, their reasoning capabilities come at the cost of high computational overhead. While adaptation via PEFT methods like Low-Rank Adaptation (LoRA) [17] offers a practical solution, it introduces other challenges in multimodal alignment.

3. Methodology

In this section, we outline the architectural frameworks and experimental variations implemented to address image captioning. To systematically analyze the transition from standard baseline architectures to state-of-the-art foundation models, our methodology parallels the structure of Section 2.

3.1. Basic Image Captioning Framework

The foundational approach to image captioning relies on an Encoder-Decoder architecture, operating under the paradigm of translating visual features into a coherent sequence of text. As illustrated in Figure 1, this framework first processes the input image through a visual encoder to extract high-level spatial or semantic feature representations, which are subsequently passed to an autoregressive recurrent decoder that generates the descriptive caption token by token, based on a text representation strategy.

To comprehensively understand the impact of individual architectural components, we evaluate diverse visual encoder backbones. Our experiments encompass established CNNs, including the ResNet family (ResNet-18, ResNet-34, and ResNet-50) [15], the VGG architecture (VGG-16 and VGG-19) [34], and EfficientNet-B0 [36]. Furthermore, we explore ViT encoders leveraging Contrastive Language-Image Pre-training (CLIP) [32], specifically evaluating the CLIP-ViT-B/32 and CLIP-ViT-L/14 variants.

For the language generation module, we compare three distinct recurrent decoder architectures: Gated Recurrent Units (GRU) [7], standard Long Short-Term Memory (LSTM) networks [16], and the more recently introduced xLSTM. To process and generate the textual data, we investigate different granularity levels for text representation, comparing character-level, subword-level, and traditional word-level tokenization strategies.

To enhance the decoder’s ability to focus on salient regions of the image during the generation of specific words, we integrate tailored attention mechanisms into the baseline architecture. For the GRU and LSTM decoders, we implement and compare soft attention and adaptive attention mechanisms, allowing the model to dynamically compute spatial attention weights over the encoder’s feature maps at each decoding time step. Conversely, for the xLSTM decoder, we utilize a *visual prefix* approach, where the pooled visual features are prepended as an initial contextual prompt to guide the textual sequence generation.

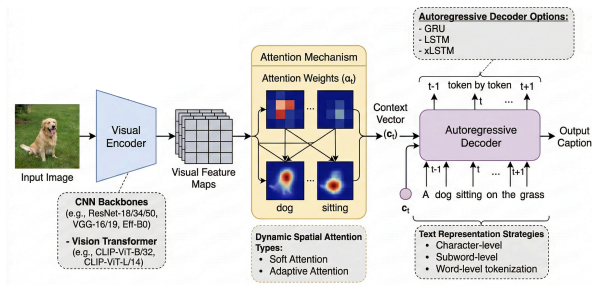


Figure 1. The Encoder-Decoder framework in image captioning.

3.2. Pretrained Vision-Language Models

While the basic CNN-RNN architectures provide a foundational baseline, they often struggle with long-range dependencies and require vast amounts of annotated data to learn visual-semantic alignments from scratch. To overcome these limitations, we transition to evaluating state-of-the-art Transformer-based architectures, leveraging the power of large-scale vision-language pre-training.

In particular, we evaluate three distinct pretrained model configurations. First, we explore two composite architectures utilizing a pretrained ViT (ViT-B/16) as the visual encoder paired with different pretrained language models for the decoder: the autoregressive GPT-2 and the bidirectional BERT. Second, we evaluate the full BLIP, which is optimized directly for unified vision-language understanding and generation.

Figure 2 shows the Multimodal Mixture of Encoder-Decoder (MED) architecture used in BLIP. This unified model operates in three distinct functional modes: (1) an unimodal encoder that processes image and text modalities separately, (2) an image-grounded text encoder that fuses

both modalities via cross-attention for multimodal understanding tasks, and (3) an image-grounded text decoder that generates text autoregressively conditioned on the extracted visual features. To enable highly efficient multi-task learning, the text encoder and decoder share all weights except for their respective self-attention layers. Specifically, the encoder employs bidirectional self-attention to build complete context, whereas the decoder utilizes causal self-attention to support autoregressive sequence generation.

A primary focus of this stage is analyzing the impact of different training and adaptation strategies. We conduct comparative experiments by first evaluating the models in their purely pretrained state (zero-shot inference) to establish a baseline of their generalized knowledge. Subsequently, we explore different fine-tuning configurations.

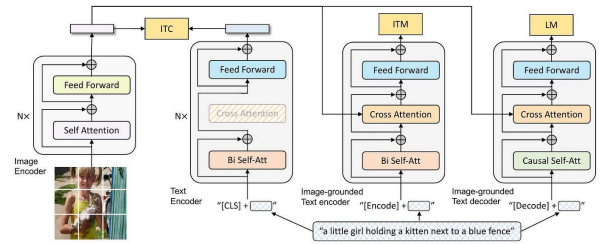


Figure 2. Multimodal Mixture of Encoder-Decoder (BLIP).

3.3. Multimodal Large Language Models

LLMs are designed as highly versatile foundational systems capable of performing a wide array of complex tasks. They can be adapted for image captioning via explicit instruction prompts, which leverages their vast pretrained semantic knowledge for visual reasoning.

In this phase, we evaluate four pretrained Qwen multimodal models (from the 2.x and 3.x series). To ensure fair comparison within our hardware limits, we select the largest feasible variant per version. Specifically, we conduct zero-shot evaluations on Qwen2-VL-7B-Instruct, Qwen2.5-VL-7B-Instruct, Qwen3-VL-8B-Instruct, and Qwen3.5-9B.

Figure 3 illustrates the foundational architecture of Qwen2-VL. The core architecture across the Qwen-VL lineage integrates three fundamental components: a ViT encoder, a Multi-Layer Perceptron (MLP) based projector, and a LLM backbone. Unlike architectures that rely on complex cross-attention mechanisms to fuse modalities, Qwen employs a straightforward MLP layer to compress and project visual features directly into the LLM’s text embedding space. This shared semantic space enables the LLM to process and *reason* over visual tokens precisely as it does with standard text tokens. Subsequent versions build upon this foundation with critical innovations:

- **Qwen2-VL [40]:** Introduces dynamic resolution to prevent detail loss, and Multimodal Rotary Position Embedding (M-RoPE) for unified 1D/2D spatial reasoning.

- **Qwen2.5-VL [2]:** Upgrades to a native dynamic-resolution ViT with Window Attention for efficiency, and significantly improves precise object grounding.
- **Qwen3-VL [3]:** Scales to 8B parameters, shifting capabilities from basic visual perception to deeper semantic reasoning in complex, ambiguous scenes.
- **Qwen3.5:** Introduces an integrated *thinking* architecture for internal deliberation, alongside a vastly expanded vocabulary for highly detailed generation.

In the second phase of our LLM exploration, we investigate the addition of LoRA to compact Qwen3.5 variants (4B, 2B, and 0.8B). As shown in Figure 4, this strategy utilizes a frozen visual BLIP encoder coupled with a projector module. We experiment with both Single Linear and MLP projectors to bridge the modality gap between the extracted visual embeddings and the compact Qwen language decoders. This comparative setup allows us to assess the adaptability of smaller LLMs to domain-specific visual captioning under strict computational constraints.

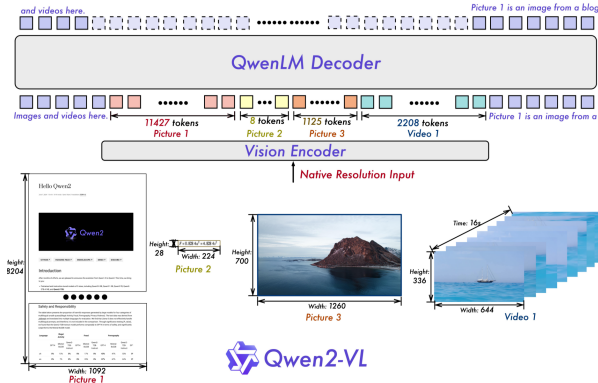


Figure 3. Core architecture of the Qwen Multimodal Models.

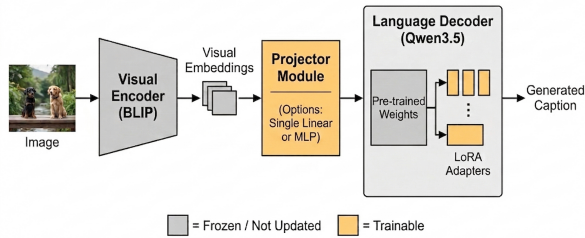


Figure 4. LoRA fine-tuning pipeline for compact Qwen models.

4. Experimental Setting

4.1. Dataset Description

We evaluate our models on the VizWiz dataset [14], a highly challenging, real-world dataset comprising images captured by visually impaired users. Unlike standard image captioning benchmarks (e.g., MS COCO [25]), VizWiz images frequently exhibit severe quality degradation, including ex-

treme blur, poor lighting, poor framing, and fingers obscuring the lens (see Figure 5). Moreover, the ground-truth text often contains conversational or noisy annotations, strictly testing the models’ domain adaptation capabilities.



Figure 5. Examples from the VizWiz dataset illustrating common challenges such as extreme blur, poor lighting, and occlusions.

For our experimental pipeline, we initially divided the provided training data into a 90% split for training (20,579 images) and a 10% development set (2,287 images) for hyperparameter tuning and model selection. Once optimal configurations were identified, the final baseline models were trained on the complete combined training set. The evaluation was conducted exclusively on the official validation set (7,542 images).

A notable characteristic of the VizWiz dataset is the variable number of ground-truth annotations per image, ranging from zero to five. To address this, images lacking any captions were first filtered out of the dataset. For the remaining data, we adopted a standard “Replicated Images” strategy. Rather than selecting a single annotation, we duplicated the image so that every available caption formed a unique image-text training pair. This approach effectively multiplies the training samples, exposing the model to a larger volume of data and more diverse sentence structures for the same visual features. During evaluation, the model’s generated prediction is compared against all available ground-truth references for that specific image, returning the highest matching score. This multi-reference evaluation ensures a much fairer assessment of performance, properly acknowledging that a single image can possess multiple, perfectly valid descriptions.

4.2. Evaluation Metrics

To comprehensively evaluate the quality of the generated captions, we employ standard automated metrics, each capturing different aspects of linguistic and semantic quality. Note that, because the poor image quality and noisy captions in the VizWiz dataset naturally limit how high automated scores can go, a model’s relative improvement is much more meaningful than its final absolute score.

- **METEOR [8]:** Used as our primary metric to drive architectural decisions, METEOR prioritizes semantic alignment—crucial for noisy VizWiz captions. It aligns exact matches, stems, and synonyms:

$$\text{METEOR} = (1 - \text{Pen}) \cdot \frac{P \cdot R}{\alpha P + (1 - \alpha)R}, \quad (1)$$

where P and R are precision and recall. A hyperparameter $\alpha \in [0, 1]$ weights recall over precision (typically ≈ 0.9), and $\text{Pen} \in [0, 1]$ is a computed penalty for fragmented matching. This focus on semantic alignment over exact matching is crucial given the noisy ground truth.

- **CIDEr [37]:** CIDEr evaluates consensus by computing the cosine similarity of TF-IDF weighted n -grams between the candidate sentence c_i and reference sentences S_i . For n -grams, it is computed as

$$\text{CIDEr}_n(c_i, S_i) = \frac{1}{m} \sum_{j=1}^m \frac{\mathbf{g}^n(c_i) \cdot \mathbf{g}^n(s_{ij})}{\|\mathbf{g}^n(c_i)\| \|\mathbf{g}^n(s_{ij})\|}, \quad (2)$$

where $\mathbf{g}^n$ represents the TF-IDF vector, s_{ij} is the reference caption j and m is the number of human references.¹

- **BLEU [30]:** Measures exact n -gram overlaps to assess structural fluency:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right). \quad (3)$$

Here, p_n is the modified n -gram precision. The weights w_n must sum to one ($\sum_{n=1}^N w_n = 1$) and are typically uniform ($w_n = 1/N$). The brevity penalty $\text{BP} \in (0, 1]$ prevents short outputs from artificially inflating scores.

- **ROUGE-L [24]:** ROUGE-L evaluates sentence structure and sequence order using the Longest Common Subsequence (LCS) between the generated sequence and references. Its F-measure is defined as

$$F_{LCS} = \frac{(1 + \beta^2) R_{LCS} P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}}, \quad (4)$$

where R_{LCS} and P_{LCS} are the LCS-based recall and precision metrics, respectively.

5. Results

5.1. Basic Image Captioning Framework

To systematically navigate the expansive combinatorial space of architectural configurations, we adopted a greedy sequential search strategy. Starting from a foundational baseline, we iteratively optimized individual modules—namely the visual encoder, language decoder, tokenization granularity, attention mechanisms, and hyperparameters. At each stage, the configuration yielding the highest METEOR score on the validation split was frozen for subsequent experiments, isolating the empirical impact of each architectural modification. A comprehensive summary of this evolutionary trajectory is detailed in Table 1.

Baseline Initialization. Our foundational architecture utilized an ImageNet-pretrained ResNet-18 vision encoder paired with a 1-layer GRU decoder (512 dimensions)

operating on character-level tokenization. This baseline achieved an initial METEOR score of 31.8.

Ablation of Visual Backbones. Leaving the decoder frozen, we evaluated various CNN (ResNet, VGG, EfficientNet) and ViT backbones. The **CLIP-ViT-B/32** emerged as the superior encoder. Because CLIP is pre-trained via contrastive learning on massive image-text pairs, its extracted visual features are inherently aligned with the semantic textual space, mitigating the modality gap.

Decoder Architecture Search. Using the frozen CLIP encoder, we swept across GRU, standard LSTM, and xLSTM architectures with varying depths and hidden dimensions. A **3-layer LSTM (1024 dimensions)** provided the optimal balance of representational capacity and stability. While deeper GRU models degraded due to vanishing gradients, the LSTM successfully managed long-term dependencies.

Tokenization Granularity. We evaluated character, subword, and word-level representations. Subword tokenization suffered from modality collapse, frequently looping fragments without properly terminating sequences. **Word-level tokenization** proved the most robust, offering direct semantic tokens that handled the noisy, colloquial nature of the VizWiz captions more effectively than character-level reconstruction.

Spatial Attention Mechanisms. To address the spatial information bottleneck inherent to global average pooling, we integrated Soft and Adaptive Attention mechanisms. **Adaptive Attention** yielded the most significant improvement. By introducing a “visual sentinel,” the model learned when to ground generated words in specific image regions (e.g., physical objects) and when to rely exclusively on its internal language model (e.g., for syntax and function words).

Hyperparameter & Decoding Optimization. Finally, we performed a Bayesian sweep over learning rates, optimizers, schedulers, and decoding strategies. The optimal configuration utilized the **Adam optimizer (lr=1e-4)** [19], a **Cosine learning rate scheduler** [27], and **Beam Search decoding (K=3)** [38]. Transitioning from greedy decoding to Beam Search globally optimized sequence coherence, pushing the final METEOR score to 38.6.

5.2. Pre-Trained Vision-Language Models

While our customized encoder-decoder architecture achieved strong performance, training models from scratch on the VizWiz dataset is inherently bottlenecked by data scarcity and severe domain noise. To overcome this, we evaluated the integration of massive pre-trained VLMs. This phase was divided into a zero-shot direct evaluation to gauge generalized multimodal alignment, followed by targeted fine-tuning strategies to adapt the representations to the VizWiz domain. A comprehensive performance

¹CIDEr was reserved primarily for final model comparisons.

Table 1. Evolution of automated metrics across the sequential optimization of the basic Encoder-Decoder framework. The table is ordered to highlight the step-by-step improvement in our primary optimization metric (METEOR). Best result per metric is highlighted in **bold**.

Experimental Stage	Selected Configuration	BLEU-1	BLEU-2	ROUGE-L	CIDEr	METEOR	Δ METEOR
0) Baseline Initialization	ResNet-18 + GRU + Char	46.4	26.2	34.1	1.1	31.8	—
1) Encoder Search	CLIP-ViT-B/32	55.5	32.6	37.5	1.8	33.8	+2.0
2) Decoder Search	LSTM (3-layer, 1024d)	55.9	33.6	38.3	2.3	35.8	+2.0
3) Text Representation	Word-level Tokenization	53.9	33.4	40.8	2.5	35.9	+0.1
4) Attention Mechanism	Adaptive Attention	58.2	37.1	42.5	3.0	37.8	+1.9
5) Hyperparameter Sweep	Adam + Cosine + Beam Search	61.4	40.2	44.0	3.6	38.6	+0.8

comparison is detailed in Table 2.

Zero-Shot Direct Evaluation. We evaluated three off-the-shelf, pre-trained configurations without any fine-tuning: a ViT-B/16 encoder paired with a GPT-2 decoder, a ViT-B/16 paired with BERT, and the unified BLIP model.

The composite models (ViT-GPT2 and ViT-BERT) struggled significantly, recording METEOR drops of -11.2 and -16.1 compared to our best Stage 1 baseline, respectively. These architectures stitch together unimodal components pre-trained on disjoint objectives, resulting in a weak cross-modal alignment space. Furthermore, ViT-BERT performed poorest overall, as BERT’s bidirectional masking objective is fundamentally ill-suited for the autoregressive nature of causal text generation.

Conversely, the BLIP model demonstrated superior zero-shot performance (32.3 METEOR). Unlike the composite models, BLIP benefits from native multimodal pre-training, utilizing joint objectives (Image-Text Contrastive, Image-Text Matching, and Language Modeling) to learn a highly aligned, shared semantic space from inception.

Domain Adaptation & Freezing Strategies. Because BLIP was pre-trained on generic, high-quality web images, its zero-shot performance still lagged behind our custom-trained baseline (-6.3 METEOR) when confronted with the extreme blur, poor lighting, and off-frame subjects typical of the VizWiz dataset. To bridge this domain gap, we explored three fine-tuning paradigms on the BLIP architecture: 1) *Frozen Vision, Tuned Text* (adapting only to the caption style), 2) *Tuned Vision, Frozen Text* (adapting only to the visual noise), and 3) *Joint Fine-Tuning* (adapting both).

Partial freezing strategies proved suboptimal. Freezing the vision encoder prevented the model from adapting its feature extraction to VizWiz’s unique visual degradations, while freezing the language decoder hindered its ability to match the linguistic structure of the ground-truth captions.

Joint Fine-Tuning yielded the best performance, surpassing our highly optimized Stage 1 baseline by +1.2 METEOR (achieving an overall score of 39.8). By fully unfreezing the architecture, BLIP successfully adapted its native multimodal representations to the VizWiz domain,

proving that adapting massive pre-trained weights is ultimately more effective than training smaller, customized architectures from scratch.

5.3. Multimodal Large Language Models

The final phase of our evaluation transitioned from traditional encoder-decoder frameworks to state-of-the-art MLLMs. These foundational models leverage massive parameter counts to perform complex visual reasoning. We divided this analysis into a zero-shot direct evaluation of the Qwen-VL lineage, followed by PEFT experiments using LoRA on compact decoder variants.

Zero-Shot Direct Evaluation. We evaluated four pre-trained models from the Qwen series: Qwen2-VL (7B), Qwen2.5-VL (7B), Qwen3-VL (8B), and Qwen3.5 (9B). To enforce concise caption generation suitable for the VizWiz dataset, we utilized a strict prompt (restricting output to under 20 words). Results are summarized in Table 3.

The results revealed an intriguing architectural paradox. The earliest and simplest model, Qwen2-VL, dominated exact-match metrics (70.6 BLEU-1 and 48.5 ROUGE-L). Its direct generation style closely aligned with the short, generic, and often noisy ground-truth captions inherent to VizWiz. In contrast, Qwen3.5 (9B) achieved the highest METEOR score (47.9). As the most advanced model, its superior vocabulary breadth excelled in semantic recall and synonym matching, accurately capturing the scene’s meaning even if the exact phrasing diverged from the ground truth. Intermediate models (Qwen2.5 and Qwen3) suffered from an architectural “middle ground,” generating descriptions too complex for high BLEU scores but lacking the deep semantic nuance of Qwen3.5 to maximize METEOR.

Parameter-Efficient Fine-Tuning (LoRA). To adapt these massive models under strict computational constraints, we utilized a frozen BLIP vision encoder paired with a trainable projector module to feed visual embeddings into various compact Qwen3.5 decoders (0.8B, 2B, and 4B parameters). We applied LoRA to the attention modules and compared a Single Linear projector against a Two-Stage MLP projector (where the MLP was warmed up prior to joint training).

Table 2. Quantitative comparison of Pre-Trained Vision-Language Models. The baseline (Best Stage 1) is provided for reference. Δ METEOR indicates the performance gap relative to the Best Stage 1 model. Best result per metric is highlighted in **bold**.

Model	Params	Encoder Tuned	Decoder Tuned	BLEU-1	BLEU-2	ROUGE-L	CIDEr	METEOR	Δ METEOR
<i>Custom Baseline (From Section 5.1)</i>									
Best Stage 1 Model	104 M	✗	✓	61.4	40.2	44.0	3.6	38.6	—
<i>Zero-Shot Direct Evaluation</i>									
ViT-B/16 + GPT2	210 M	✗	✗	51.8	31.9	37.1	1.9	27.4	-11.2
ViT-B/16 + BERT	196 M	✗	✗	39.8	17.0	27.1	0.3	22.5	-16.1
BLIP (Zero-Shot)	247 M	✗	✗	55.2	37.4	42.5	3.3	32.3	-6.3
<i>BLIP Domain Adaptation (Fine-Tuning)</i>									
BLIP (Vision Frozen)	247 M	✗	✓	59.9	38.2	42.4	3.3	38.6	0.0
BLIP (Text Frozen)	247 M	✓	✗	58.0	39.3	44.0	3.7	37.6	-1.0
BLIP (Joint Fine-Tuning)	247 M	✓	✓	61.4	40.2	44.0	3.9	39.8	+1.2

As shown in Table 4, the Single Linear projector unexpectedly outperformed the MLP across all model scales, with the Qwen3.5-4B variant achieving the highest METEOR (27.9). However, compared to our previous experiments, performance plummeted severely. This catastrophic drop indicates severe “modality collapse”. Because the BLIP visual features and Qwen text embeddings originate from entirely disjoint latent spaces, the initial projector outputs appear as out-of-distribution to the LLM. Consequently, it learns to permanently ignore the image tokens, relying entirely on its language priors to hallucinate syntactically correct but visually irrelevant captions (as explicitly demonstrated in the qualitative analysis in Figure 6).

Table 3. Zero-shot direct evaluation of the Qwen-VL family. The simplest model maximizes exact-match metrics, while the most advanced model (Qwen3.5) maximizes semantic recall (METEOR). Best result per metric is highlighted in **bold**.

Model	BLEU-1	BLEU-2	ROUGE-L	CIDEr	METEOR
Qwen2-VL (7B)	70.6	48.2	48.5	6.0	45.5
Qwen2.5-VL (7B)	55.0	33.4	40.0	3.8	40.7
Qwen3-VL (8B)	56.2	32.9	38.4	4.5	39.2
Qwen3.5 (9B)	61.0	39.7	44.9	5.0	47.9

Table 4. METEOR scores for LoRA fine-tuning of compact Qwen3.5 decoders using a frozen BLIP encoder. The low scores indicate modality collapse. Best result is highlighted in **bold**.

Decoder Model	Single Linear Projector	MLP Projector
Qwen3.5-0.8B	27.8	20.4
Qwen3.5-2B	27.1	19.6
Qwen3.5-4B	27.9	22.9

5.4. Qualitative Analysis

To contextualize the quantitative metrics and understand the true impact of our architectural upgrades, we conducted a

qualitative analysis on the VizWiz validation set. Figure 6 illustrates the evolution of our models across two challenging real-world scenarios: an off-angle shot of a highly reflective laptop screen, and a photograph of a computer monitor displaying an image.

The Hallucination Baseline. Our initial Stage 1 baseline (ResNet-18 + GRU) failed entirely on both examples. Because it lacked pre-trained semantic alignment and spatial attention, it hallucinated nonsensical objects, identifying a “black and white striped shirt” in both images.

Incremental Semantic Refinement. As we upgraded the architecture (CLIP encoder, Adaptive Attention, and hyperparameters), the Final Stage 1 model successfully identified the “computer screen” and “house,” though its descriptions remained generic. The introduction of the pre-trained BLIP model further refined the context. For instance, the fine-tuned BLIP correctly identified the “blue screen and yellow box”, whereas its zero-shot version hallucinated a “meteor” due to the intense glare on the screen. However, neither the custom frameworks nor BLIP possessed the necessary OCR capabilities to read the specific text on the screen.

Semantic Depth vs. Global Context. While Qwen3.5 (9B) possesses superior semantic vocabulary and fine-grained object recognition—evidenced by its ability to accurately read “Toshiba” and “Safe Mode” in the first image, and identifying the abandoned house with overgrown vegetation in the second image—it can sometimes over-focus on internal contents at the expense of the broader physical medium. Note that, for the second image, Qwen3.5 completely misses that the scene is displayed on a computer monitor, describing it as a physical street scene. In contrast, the fine-tuned BLIP model correctly captures the global context.

Visualizing Modality Collapse (Qwen3.5 4B LoRA). The final captions in Figure 6 visually confirm the modality collapse experienced during our LoRA fine-tuning exper-

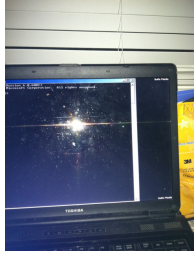


Image 1

Ground Truth Captions:

- here is a picture of a laptop screen with safe mode.
- a Toshiba laptop in safe mode with command prompt open.

Baseline (Stage 1): A black computer monitor with a black and white striped shirt and a black background.

Final Model (Stage 1): a computer screen with a blue background and white text is shown.

BLIP (Pre-Trained): a laptop computer sitting on a desk with a screen showing a meteor.

BLIP (Fine-Tuned): a laptop computer with a blue screen and a yellow box in the background.

Qwen3.5 (Pre-Trained 9B): A Toshiba laptop displaying a Microsoft command prompt with "Safe Mode" visible on the screen.

Qwen3.5 (LoRA Fine-Tuned 4B): The image displays a close-up of a dark glass bottle, likely containing wine or beer... **X(Collapse)**

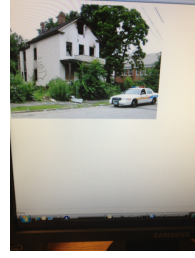


Image 2

Ground Truth Captions:

- A house with some police car in the monitor window.
- computer display monitor displaying old home and police vehicle.

Baseline (Stage 1): A computer screen with a black and white striped shirt and a black background.

Final Model (Stage 1): a computer screen with a picture of a house and trees.

BLIP (Pre-Trained): a computer screen with a picture of a house.

BLIP (Fine-Tuned): a computer screen showing a picture of a house and a car.

Qwen3.5 (Pre-Trained 9B): A police car is parked in front of an abandoned house with overgrown vegetation.

Qwen3.5 (LoRA Fine-Tuned 4B): The image displays a close-up of a dark, textured fabric, likely a fleece or wool blanket, with a subtle pattern of small dots. **X(Collapse)**

Figure 6. Qualitative analysis of the evolution of our captioning models. The results demonstrate the transition from hallucinating baselines, to generic intermediate models and the highly specific Qwen3.5 (9B). Last two rows illustrate the severe modality collapse experienced during the LoRA fine-tuning of the 4B model.

iments (Section 5.3). When asked to describe the laptop screen, the fine-tuned 4B model confidently hallucinated a “dark glass bottle, likely containing wine or beer.” For the police car image, it described a “fleece or wool blanket.” This proves that the LLM decoder learned to completely ignore the noisy outputs from the linear projector.

6. Conclusions

In this work, we presented a systematic evaluation of image captioning architectures on the highly challenging VizWiz dataset. Our investigation spanned from optimizing foundational Encoder-Decoder frameworks, analyzing the impact of visual backbones, recurrent decoders, and attention mechanisms to adapting massive pre-trained Vision-Language Models and state-of-the-art Multimodal LLMs.

Our experiments yielded several critical insights into multimodal representation learning. First, within basic frameworks, utilizing semantically aligned visual features and dynamic visual grounding proved paramount for robust performance. Second, when transitioning to pre-trained models, we found that joint fine-tuning of both the encoder and decoder is absolutely essential. Without this end-to-end adaptation, models cannot overcome the severe domain shift introduced by VizWiz’s extreme blur, poor lighting,

and conversational ground-truth annotations. Third, our evaluation of the Qwen-VL series demonstrated massive MLLMs excel in zero-shot environments. Qwen3.5 (9B) achieved the highest METEOR by maximizing semantic recall and leveraging its vast vocabulary, although qualitative analysis revealed it can occasionally over-focus on fine-grained objects at the expense of global physical context.

Finally, our Parameter-Efficient Fine-Tuning (LoRA) experiments exposed a base architectural challenge: modality collapse. Relying on a simple linear projector to map frozen visual features into an LLM’s text space failed to bridge the representational gap. Consequently, the decoder ignored the visual tokens and defaulted to acting as a pure language model—generating captions by simply predicting the most statistically probable sequence of words based on its pre-training, effectively hallucinating syntactically fluent but visually irrelevant text.

Future work should address this alignment bottleneck. Promising extensions include exploring joint end-to-end training paradigms where the vision encoder is unfrozen alongside the LLM, allowing the visual representations to dynamically adapt to the text space. Additionally, replacing simple linear projections with sophisticated cross-attention mechanisms could foster deeper, more robust alignment.

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